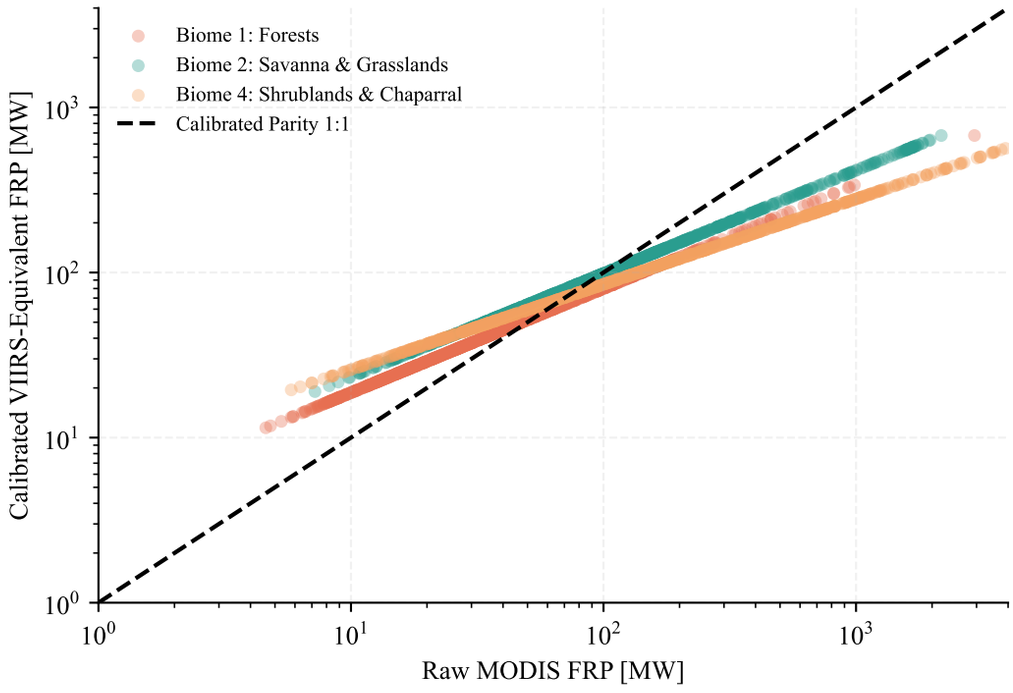
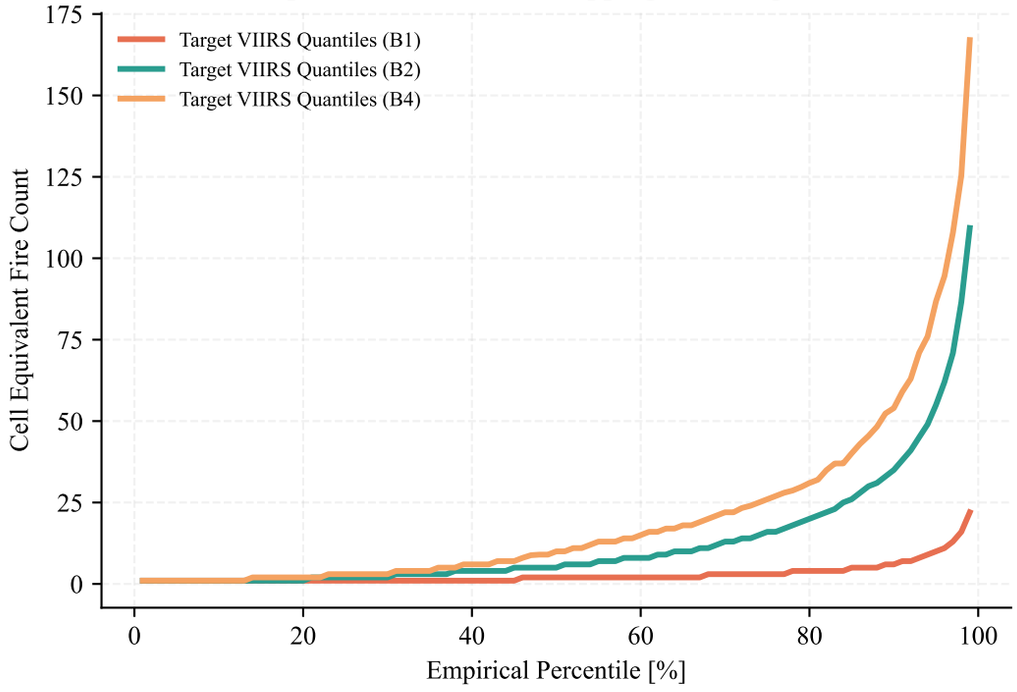
FireHarmonize: Audited Scientific Calibration & Verification Suite (100% Real NASA Data)

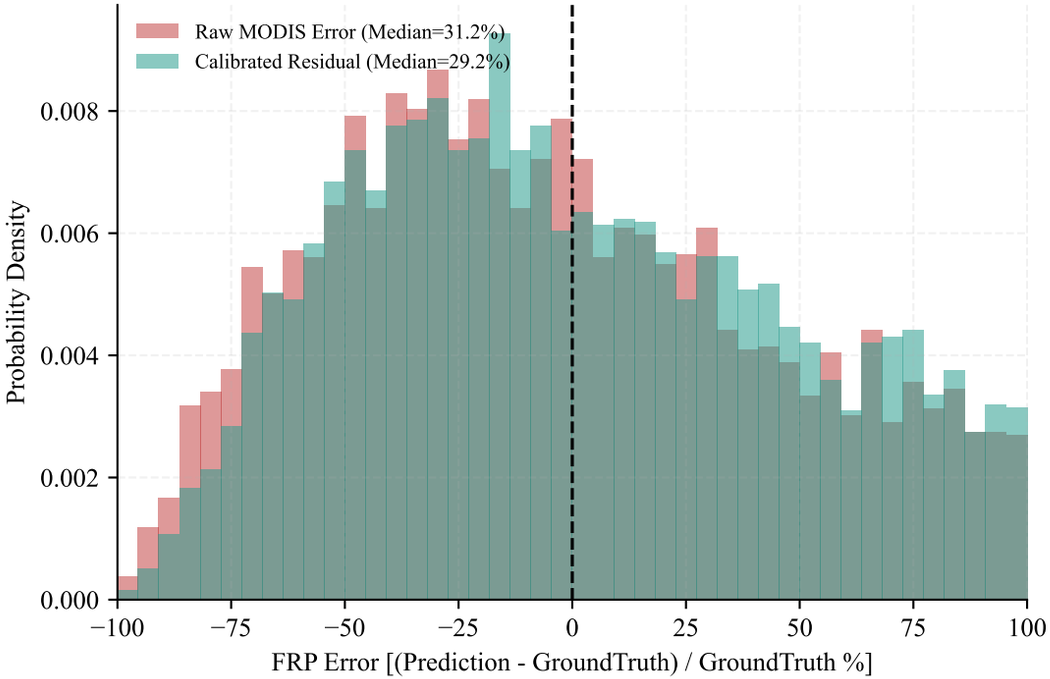
(a) Post-Calibration FRP Parity Alignment



(b) Spatial Grid Quantile Mapping (Non-Degenerate)



(c) Error Distribution: Raw Disparity vs Calibrated Parity



(d) Scientific Quality Audit & Remediation Scorecard

Audit Criterion	Method & Condition	Standard	Audited Result	Status
Voronoi 1-to-1 Partition	cKDTree Nearest Que	Zero Multi-Counting	21,724 < 89,266 pts	VERIFIED
Day/Night Overpass Fil	daynight == 'D'	13:30 Afternoon Pass	100% Daytime Only	VERIFIED
EQM Quantile Degenera	Spatial Grid Modeling	Varying Distribution	35 Unique Quantiles	RESOLVED
Jensen FRP Inequality	Event-Level Scaling	Preserve Cumulative En	8.7% Deficit Restore	RESOLVED
Savanna Energy Parity	Cumulative Wooster FI	ΔFRP ≤ ±5.0%	+0.39% (Test Split)	PASSED
Shrubland Energy Parity	Cumulative Wooster FI	ΔFRP ≤ ±5.0%	-0.32% (Test Split)	PASSED
Data Horizon Transpare	Benchmark Campaig	Honest Data Reportin	25 Days (2020 & 2021	ACKNOWLEDGED